

EDUCATION

Ph.D. in Climate Science
08/2018 – 04/2022

Utah State University
Advisor: Shih-Yu Simon Wang
Dissertation: *Impacts of the Changing Pacific on North American Drought, Atmospheric Rivers, and Explosive Cyclones*

B.S. in Environmental Science
08/2012 – 05/2016

Loyola Marymount University
Focus on Earth Science

RELEVANT PROFESSIONAL EXPERIENCE

Postdoctoral Associate
08/2022 – present

Yale University

Graduate Research Assistant
08/2018 – 07/2022

Utah State University

Teacher's Assistant
01/2019 – 05/2021

Utah State University
Meteorology, Climate and Climate Change

Research Assistant
05/2019 – 06/2020

N.O.A.A. Climate Prediction Center
Empirical prediction of wintertime teleconnections

Field Guide and Research Assistant
06/2016 – 09/2016

Campanario Biological Station
Tropical ecology guide and educator for student groups and tourists

PEER-REVIEWED PUBLICATIONS

IN PREPARATION:

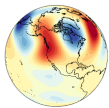
Stuivenvolt-Allen, J., Fedorov, A. (2023). Impacts of changes in the structure of wind anomalies on ENSO statistics under global warming. *Journal of Climate*.

Stuivenvolt-Allen, J., Chikamoto, Y., Wang, S.-Y. (2023). Emergence of decadal hydroclimate regime in the western United States. *Environmental Research Letters*.

Garuba, O., Liu, W., **Stuivenvolt-Allen, J.** (2023). The role of oceanic and atmospheric feedbacks in the response of the Atlantic Meridional Overturning Circulation to a CO₂ increase. *Journal of Climate*.

PUBLISHED:

10. Johnson, Z., **Stuivenvolt-Allen, J.**, Mahan, H., Meyer, J., Miksch. (2023). Upper Colorado River streamflow dependencies on summertime synoptic circulations and hydroclimate variability. *Journal of Hydrometeorology*. <https://doi.org/10.1175/JHM-D-23-0053.1>



9. Dalanhesse, L., **Stuivenvolt-Allen, J.**, LaPlante, M., Wang, S.-Y. (2023). An inter-comparison of South American extratropical cyclogenesis using Brazilian Navy data and ERA5: *International Journal of Climatology*. <https://doi.org/10.1002/joc.8251>
8. **Stuivenvolt-Allen, J.**, Wang, S.-Y., Chikamoto, Y., Meyer, J.D., Johnson, Z.F. and Deng, L. (2023). Growing Pacific linkage With Western North Atlantic Explosive Cyclogenesis: *Journal of Climate*. <https://doi.org/10.1175/JCLI-D-22-0784.1>
7. Wright, T., **Stuivenvolt-Allen, J.**, Afram, G., Hasan A., N., Ratterman, C., Zhang, W. (2023). Responses of extreme discharge to changes in surface air and dewpoint temperatures in Utah: Seasonality and Mechanisms. *Water*. <https://doi.org/10.3390/w15040688>
6. Pokharel, B., Sharma, S., **Stuivenvolt-Allen, J.**, Wang, S.-Y. (Simon), LaPlante, M., Gillies, R. R., Khanal, S., Wehner, M. F., Rhoades, A. M., Hamal, K., Liu, W.-Y., Mukherjee, S., & Aryal, D. (2023). Amplified drought trends in Nepal increase the potential for Himalayan wildfires. *Climatic Change*. <https://doi.org/10.1007/s10584-023-03495-3>
5. **Stuivenvolt-Allen, J.**, Wang, S.-Y. (2022) North American fire weather catalyzed by the extratropical transition of tropical cyclones: *Climate Dynamics*. <https://doi.org/10.1007/s00382-022-06561-1>
4. **Stuivenvolt-Allen, J.**, Wang, S.-Y. S., Johnson, Z., & Chikamoto, Y. (2021). Atmospheric rivers impacting Northern California exhibit a quasi-decadal frequency. *Journal of Geophysical Research: Atmospheres*, 126, e2020JD034196. <https://doi.org/10.1029/2020JD034196>
3. Morgan, B., Spangler, K., **Stuivenvolt-Allen, J.**, Morrisett, C. N., Brunson, M. W., Wang, S.-Y. S., & Huntly, N. (2021). Water Availability for Cannabis in Northern California: Intersections of Climate, Policy, and Public Discourse. *Water*, 13(1). <https://doi.org/10.3390/w13010005>
2. **Stuivenvolt-Allen, J.**, Wang, S.-Y. S., LaPlante, M. D., & Yoon, J.-H. (2021). Three Western Pacific Typhoons Strengthened Fire Weather in the Recent Northwest U.S. Conflagration. *Geophysical Research Letters*, 48(3), e2020GL091430. <https://doi.org/10.1029/2020GL091430>
1. **Stuivenvolt-Allen, J.**, & Wang, S.-Y. S. (2019). Data-mining climate variability as an indicator of U.S. natural gas: *Frontiers in Big Data: Integrating Big Data with Earth System Models of Natural and Human Systems*. <https://doi.org/10.3389/fdata.2019.00020>

RESEARCH INTERESTS

El Niño Southern Oscillation

Westerly wind bursts and ENSO development

Role of WWBs and atmospheric forcing in driving super-Niño events in the warming world

Hydroclimate

Atmospheric Rivers and Drought

Low-frequency variability and dynamics of atmospheric rivers and multi-year droughts in the western US

Climate Dynamics

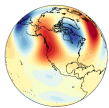
Ocean-atmosphere interactions

Impacts of ENSO and tropical teleconnections on North American weather and climate

Fire Weather

Typhoons to fires

Responses of North American fire weather to perturbations associated with western North Pacific tropical cyclones



Climate Prediction

Empirical prediction

Prediction of winter teleconnections at 1-2 months lead time. Statistical prediction tools for Nepal fire occurrences from meteorological variables

Coupled climate models

CESM and E3SM

Capable of running the Community Earth System Model and the Energy Exascale Earth System Model with extensive source code modifications

Idealized models

ISCA climate model

Experience in idealized simulations with a wide range of parameterizations and simplified assumptions

FELLOWSHIPS

Climate Adaptation Science Fellowship
2018-2020

NSF Research Traineeship

Research topic: Intersections of Climate, Policy, and Public Discourse in Northern California

Southwest Climate Adaptation Science Fellowship
2020-2021

SW Climate Adaptation Science Center

Research topic: Landscape regeneration in a changing climate after large and destructive fires.

Campbell Scientific Graduate Fellowship
2019 and 2020

AGRI Apogee Instruments

Awarded to a student in the Plants, Soils, and Climate department for quality, hypothesis driven research.

SERVICE AND OUTREACH

Seminar Coordinator

Atmospheric, Ocean, and Climate Dynamics

Peer Review

Journals

Environmental Research Letters, Geophysical Research Letters, Journal of Climate, Frontiers in Earth Science

Scientific Journalism

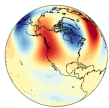
[Improving Our Relationship With Fire](#)

[The Butterfly and the Blaze – Eugene Weekly](#)

[Utah Water Issues](#)

Weather forecasting
2018-2022

Utah Public Radio



**Ignatians Service
Organization**
2012-2016

Community Service in Los Angeles

CONFERENCE PRESENTATIONS AND WORKSHOPS

CESM Workshop 2023

Oral: Wind stress contributions to changes in ENSO statistics from climate change

**European Geophysical Union
2023**

Oral: Growing Pacific Linkage with Explosive Cyclogenesis

**Yosemite Hydroclimate
Workshop 2021**

Oral: *Long-term cycles of water availability and atmospheric rivers in Northern California*

Oral: *Connecting extremes: Typhoons catalyze fire weather in the western United States*

**American Geophysical Union
2020**

Poster: *Water availability for cannabis in Northern California: Intersections of Climate, Policy, and Public Discourse*

**American Meteorological
Society Annual Meeting 2019**

Oral: *The modulation of natural gas through winter climate and cyclone variability*

**Southeast Asian Paleo
Environments Consortium
2020**

Oral: *Climate dynamics and climate data*

Workshop: *Climate science, data analysis, and stewardship*

**NSF Research Traineeship
Annual Meeting 2020**

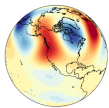
Poster: *Partnerships to enable science that is useful and used*

**Southern California Academy
of Science 2015**

Poster: *Measuring runoff fecal indicator bacteria removal efficiency in the Ballona Creek rain garden*

AWARDS AND CERTIFICATIONS

2022	Doctoral Researcher of the Year: Plants, Soils, and Climate Department
2021	Top Award for Science Writing from the American Society of Journalists and Authors: The Butterfly and the Blaze
2020	Graduate Enhancement Award
2019	Best Research Presentation: Plant, Soils and Climate Departmental Showcase
2017	Utah Master Naturalist: Wetlands Certification
2017	Utah Master Naturalist: Mountains Certification



JACOB STUIVENVOLT-ALLEN

Postdoctoral research associate | Yale

Curriculum Vitae

Jacob.stuivenvolt-allen@yale.edu | [Website](#)

2016

Sigma Xi National Research Honor Society Member